Validated, yet Wrong: Four Ways Small-n Metabolomics Machine Learning Passes Standard Checks

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Abstract—Small-cohort metabolomics studies routinely support their headline scores with cross-validation, permutation testing, and feature-attribution checks. We audit four of our own earlier analyses of public NMR data and show four failure modes in which a standard check passes while the reported result is wrong or uninterpretable: (i) an outcome manufactured from the features through change scores, (ii) putatively exogenous compounds that a feature-removal faithfulness test ranks as highly as endogenous metabolites, (iii) a label perfectly aliased with the sample-collection protocol, and (iv) a label whose name does not match what it encodes. For each, we give the check it passes and a targeted control that exposes it. In a bariatric-surgery responder forecast, a label-permutation test with refitting gave $p = 0.001$, yet the model scored below a learn-nothing baseline, and after correcting the outcome the trained model (PR-AUC 0.554) was still no better than that baseline (0.589). In simulation with no forecastable individual response, label permutation rejected in 11–100% of datasets at $\alpha = 0.05$, rising with sample size; residualising change on each marker's own baseline still rejected in 8–29%, whereas residualising on all baselines jointly held the rate at 5–8%, slightly above nominal. We release the controls as a reusable checklist with reference code.

Index Terms—metabolomics, NMR, validation, data leakage, regression to the mean, permutation test, feature attribution, confounding, reproducibility

I. Introduction

Machine-learning studies on small metabolomics cohorts ($n < 200$) can report near-perfect discrimination, and readers are asked to trust these numbers because the authors used cross-validation, a permutation test, and an explanation method such as SHAP [12]. Yet leakage and related evaluation errors have been documented across hundreds of ML-based studies in many fields [8], and omics data add hazards of their own: acquisition batch, sample handling, and repeated measurements of the same person [9]. With a few dozen subjects, any of these can produce a near-perfect score.

Each of these checks answers a narrow question. Cross-validation asks whether the model generalises to new samples drawn the same way; a label permutation asks whether the features carry any association with the label; a faithfulness test asks whether the model relies on a feature. None asks whether the label measures what it claims to, or whether the association is biological. A result can therefore pass any one of these checks and still be wrong.

The four analyses we examine are our own earlier work: a responder forecast and a feature-attribution study on MTBLS242, and a diabetes classifier on MTBLS1 [19]–[21]. Each passed the checks we had applied at the time. This paper makes four contributions:

1) A taxonomy of four failure modes, each paired with the standard check it passes and a targeted control that catches it (Table I).
2) A pairing-destroyed null for change-score outcomes, a conservative diagnostic for outcomes manufactured by regression to the mean.
3) A simulation study (36,000 datasets) showing that label permutation becomes more wrong as n grows, that the common per-marker regression-to-the-mean correction is itself liberal for correlated markers, and that joint residualisation brings the error rate close to nominal.
4) An open checklist and reference implementations of the controls for metabolomics ML studies.

II. Related Work

A. Regression to the mean and mathematical coupling

Correlating change with initial value is a known statistical artefact [5]–[7], and a change score that shares terms with the baseline is mathematically coupled to it

Code, simulation, fixed analysis designs and verification scripts: https://github.com/Calvinkub/validated-yet-wrong (tag paper-v1).

[15]: measurement error and within-person variation make extreme baselines move back toward the mean, so baseline "predicts" change even when nothing happens. The epidemiological remedies (analysis of covariance, Oldham's method) are well known [16], [17]. Our contribution is not the artefact but its interaction with ML practice: an outcome defined from change scores, a flexible model given the baselines as features, and a label permutation as the significance test. We show that this combination certifies the artefact, and that the per-marker correction a careful analyst might add is not sufficient.

B. Leakage and confounding in ML-based science

Kapoor and Narayanan [8] catalogue leakage types, including features that act as proxies for the outcome and non-independence between training and test data. Batch effects can be perfectly confounded with the biological variable of interest, in which case no analysis can separate them [9]; analytical and pre-analytical bias in serum metabolomics is well documented [18], [23], [27], and exogenous compounds introduced during sampling can appear in NMR spectra [26]. Reporting guidelines such as TRIPOD+AI [10] require that the outcome definition and its relation to the predictors be stated. Our cases are concrete metabolomics instances of these categories, each paired with a test that detects it.

C. Feature-attribution faithfulness

SHAP [12] attributes a prediction to features; ROAR [11] tests whether the attributed features matter by removing them and retraining. Both answer a question about the model, and attribution methods can mislead even about model behaviour [24], [25]. ROAR was not designed to say whether a feature is biological, but a low ROAR curve is easily read as confirming that the top-ranked metabolites are meaningful. We use putatively exogenous compounds and a synthetic artefact to show why that reading is unsafe.

D. Permutation tests for classifiers

A label permutation tests the null hypothesis that features and labels are independent [13]. Rejecting it shows only that some association exists. If the label is computed from the features, that association is guaranteed, and the test answers a question nobody asked. The permutation must also refit the model on every permuted label vector: permuting labels against fixed out-of-fold scores is not a valid test (Section VIII); the broader fragility of resampling-based validation on small cohorts is well established [28]–[31].

III. Data and Common Setup

A. Datasets

MTBLS242 [1], [2]: serum ^{1}H-NMR from a bariatric surgery cohort, released 10 Feb. 2017. The sample sheet lists 465 samples at five visits, labelled in its time-point factor as preoperative (106), and 3, 6, 9 and 12 months after surgery (98, 98, 92 and 71); 21 metabolites are quantified. We re-downloaded the sheet from MetaboLights on 15 Oct. 2026 and it is byte-identical to the copy used in all experiments. Patient identifiers are not a column; they are recoverable from sample names with the pattern `^(\d+)[-_](\d+)[-_]S$` (one 6-month sample, 2_0242_2, does not match and does not affect any analysis here, which uses only the preoperative and 12-month visits). All 71 twelve-month patients also have a preoperative sample. The sheet records neither the surgical procedure per sample nor any clinical covariate such as body-mass index, although the study design lists both gastric bypass and sleeve gastrectomy.

MTBLS1 [3]: urine ^{1}H-NMR, type 2 diabetes vs. control; 132 samples from 40–42 subjects (12 volunteers sampled at seven time points; 28–30 patients whose sample-to-patient membership is not deposited; see Section VI).

MTBLS881 [32]: serum ^{1}H-NMR from 50 adults undergoing cardiac surgery with cardiopulmonary bypass, sampled at six times (from -1 to 20 h) from two sites (left atrium, LA; pulmonary artery, PA). We use it only to replicate Case 1 on a second cohort: 47 patients have both the -1 h and 20 h samples at each site.

B. Models and evaluation

We deliberately reuse the models from the original analyses rather than tuning new ones, since the point is to evaluate the checks, not the models. Cases 2–4 use XGBoost [14] with monotonic constraints whose directions were assigned from the metabolomics literature. Case 1 uses L2-penalised logistic regression with literature-assigned sign constraints on the marker coefficients; its replication (Case 1b) uses the same model without constraints. The headline metric is PR-AUC (average precision), reported next to the prevalence it must beat. Cross-validation is repeated and grouped by subject wherever subjects can be identified. Every analysis of Case 1 includes a learn-nothing baseline: a fixed, unfitted score that any trained model must beat, with intervals from bootstrap resampling of subjects.

IV. Case 1: An Outcome Manufactured from the Features

A. Setup

Task: predict from the preoperative profile whether a patient is a twelve-month "strong responder," defined as $\geq$9 of 11 markers changing in the expected direction. 71 paired patients, 31 responders. The 11 markers and their expected directions were fixed from the obesity and insulin-resistance literature [33], [34], not from these data: expected higher in obesity are L-valine, L-leucine, L-allo-isoleucine, L-tyrosine, D-phenylalanine, L-alanine, lipoproteins and lactate; expected lower are glycine, L-glutamine and histidine. The other ten of the 21 metabolites carry no direction. Three of them (isopropanol,

TABLE I
Four failure modes: the standard check each passes and the control that exposes it. "Reported result" is the score of our original analysis.

Case	Failure mode	Data	Reported result	Standard check passed	Exposing control
1	Outcome manufactured from features	MTBLS242	PR-AUC 0.683	Label permutation (refit), $p = 0.001$	Learn-nothing baseline; joint residualisation
2	Putative contaminants ranked with biology	MTBLS242	PR-AUC 0.973	ROAR faithfulness	Known-artefact positive control
3	Collection protocol equals label	MTBLS1	PR-AUC 0.976	Subject-grouped CV	Metadata and protocol audit
4	Label name ≠ label meaning	MTBLS242	ROC-AUC 0.960	Subject-grouped CV	Label-provenance audit

methanol, dimethyl sulfone) are the putatively exogenous compounds of Case 2, dropped in the corrected designs (18 features remain).

B. Why the standard check passes

Baseline x_0 is both the predictor and a term in the outcome $\Delta = x_{12} - x_0$. Under pure noise, $\mathrm{corr}(x_0, \Delta) < 0$ by construction; the observed median was -0.62. A label permutation breaks the label–feature link entirely, so it tests against a null that the coupled outcome trivially beats.

C. Exposing controls

Learn-nothing baseline. A fixed score built from the baselines and the expected marker directions, with no fitting, exploits regression to the mean as well as any trained model can. A trained model that does not beat it has learned nothing beyond the coupling.

Pairing-destroyed null. Shuffle which twelve-month sample belongs to which subject, recompute the outcome from scratch, and rescore. This keeps the arithmetic coupling and removes any real individual-level signal. Because shuffling breaks the positive correlation between visits, the null has stronger coupling than the observed data, so its lift is expected to exceed the observed lift even when a real response exists (Section VIII). It is therefore a one-sided diagnostic: a model that clears it has shown something beyond regression to the mean; a model that does not clear it has shown nothing either way.

Joint residualisation. Inside each training fold, regress each marker's change on all 11 baselines and define the label from the residuals. Section VIII shows that this, unlike per-marker residualisation, keeps the false-positive rate close to nominal.

D. Result

Original design (raw change, ≥9 of 11 markers, putative contaminants included). The trained model reaches PR-AUC 0.683 against a prevalence of 0.437 (our originally reported 0.671 does not reproduce exactly on re-run). A label permutation with the model refitted on every permutation (1,000 permutations) gives $p = 0.001$, the floor. Yet the learn-nothing scorer reaches 0.709, above the model (gap -0.025, repeat-level interval $[-0.047, -0.006]$). Consistent with this, the observed lift (1.57×) does not clear the pairing-destroyed null (median 2.04×; Fig. 1), and refitting the full model on each shuffle gives the same picture (median 1.99×, 500 replicates). The permutation test certifies a model that does not beat an unfitted score.

TABLE II
Case 1, corrected designs. PR-AUC values are means over 20 repeats of the per-fold average precision. Gap = model − learn-nothing scorer, computed on held-out predictions pooled within each repeat, which is why it differs slightly from the difference of the PR-AUC rows; interval from a bootstrap over the 71 subjects. Prevalence differs because the joint design yields a different responder count, so compare lifts and gaps, not PR-AUC levels.

	Per-marker resid.	Joint resid.
Responders / 71	31	37
Prevalence	0.437	0.521
Model PR-AUC	0.483	0.554
Learn-nothing PR-AUC	0.522	0.589
Gap (pooled)	−0.042	−0.037
95% interval	[−0.149, +0.052]	[−0.111, +0.037]
Refit permutation p (B)	1.000 (200)	1.000 (1000)
Model lift	1.11	1.06
Pairing-null lift, median	1.04	1.05

Corrected designs (change residualised inside each training fold, contaminants removed). Table II shows the per-marker design of the original repository and the joint design recommended by Section VIII. In both, the trained model does not beat the learn-nothing scorer, and the subject-bootstrap interval for the gap spans zero. The refitted permutation test also finds nothing, but its null sits above the observed statistic in every replicate ($p = 1.0$), which a well-calibrated null would rarely do. A plausible cause, which we did not test, is that the sign constraints force a literature-signed score even on permuted labels. We therefore do not read $p = 1.0$ as a calibrated p-value. The defensible statement is that the model is no better than the scorer; at $n = 71$ the interval cannot exclude a modest real gain.

E. Replication on a second cohort

We repeated the four tests on MTBLS881 with the design fixed in a pre-registration file before any model was fitted: baseline −1 h versus 20 h; both sites analysed

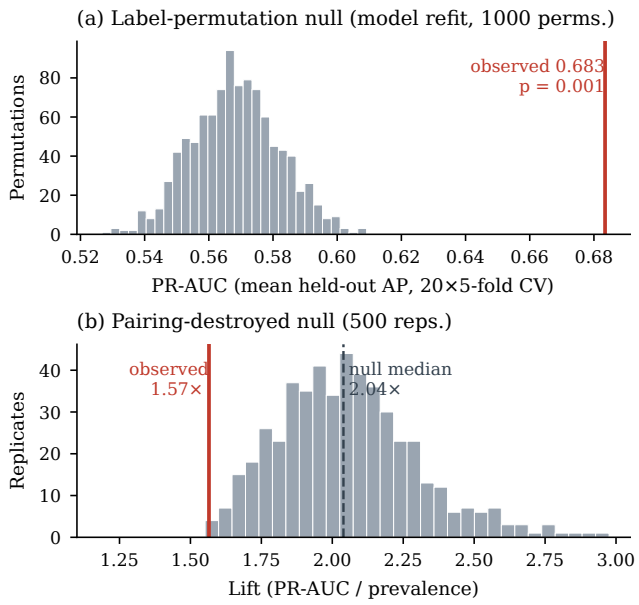


Fig. 1. Original (withdrawn) design. (a) Label-permutation null of average precision, model refitted per permutation. (b) Pairing-destroyed null of lift. The label permutation rejects. The pairing-destroyed null sits above the observed model, which is expected by construction (Section VIII) and means only that the model has not shown anything beyond regression to the mean.

separately (LA declared primary); 11 marker metabolites chosen in advance; improvement direction taken from training-fold mean change only; K chosen so prevalence is near 0.5; exogenous drugs (propofol, heparin, ethanol) excluded. The model is unconstrained ridge logistic regression with the same repeated CV as Case 1. One deviation from the pre-registration: after a single timing run, which printed K, prevalence and PR-AUC for one site, we reduced the number of null replicates for run time (300 permutations for the main designs, 200 for the exogenous-compound variant, 150 refitted pairing replicates); designs, markers and the K rule were not changed. This is a different, acute setting (hours after surgery, not a year), so it tests the statistical artefact, not the surgical biology. With 47 patients every interval is wide (Table III).

Raw change replicates. At both sites a label permutation rejects ($p = 0.007$), yet the model does not beat the learn-nothing scorer (pooled gap -0.037 and $+0.013$, both intervals spanning zero), and the pairing-destroyed null reaches equal or higher lift (median 1.49 and 1.40 vs. observed 1.29 and 1.19). Refitting the full model on each shuffle puts the observed lift inside the null ($p = 0.86$ at LA, 0.91 at PA).

Residualised outcomes replicate only partly. At PA, joint residualisation shows no signal ($p = 0.977$), matching MTBLS242. At LA the per-marker outcome is permutation-significant ($p = 0.003$), which the simulation predicts can happen without real signal. For LA joint residualisation the permutation test is not significant ($p = 0.103$) while the model's lift (1.45) exceeds the scorer-based pairing-null 95th percentile (1.27). The tests disagree here, and 47 patients cannot resolve it. Every gap interval in Table III includes zero.

Exogenous compounds create apparent signal. In a variant that adds propofol, heparin and ethanol both as features and as markers (14 markers, so the K rule now selects $K = 8$ for the joint outcome), the joint-residualised outcome becomes permutation-significant at both sites ($p = 0.005$, the floor for 200 permutations), against $p = 0.103$ and 0.977 without them. At LA the gap also becomes $+0.163$ [$+0.069$, $+0.252$], the only interval in this replication that excludes zero; at PA the gap stays near zero ($+0.008$). Because the variant changes both the features and the label definition, it does not isolate a mechanism, but the pattern matches Case 2: drugs administered during surgery can distinguish samples for reasons unrelated to the response being modelled.

V. Case 2: Putative Contaminants Ranked with Biology

A. Setup

ROAR [11] removes the top-k attributed features, retrains, and compares the score with removing k random features. Here the attribution is SHAP on the MTBLS242 pre- vs. 12-month task (177 samples, 21 metabolites). We repeat our original analysis with three fixes: cross-validation is grouped by patient (10 repeats of 5 folds), attribution is recomputed after every retrain (recursive ROAR), and the random null uses 500 distinct draws per k (exhaustive for $k \leq 2$) under the same folds as the informed runs. Grouped baseline PR-AUC is 0.973; ungrouped CV gives 0.972. Table IV lists all 21 features, the two trios compared below and the order in which recursive ROAR removes them.

B. Why the standard check passes

Recursive ROAR removes, in order, dimethyl sulfone, isopropanol, L-valine, glycine, methanol, and then further metabolites. From $k = 3$ onward the curve lies below almost the entire random band (null percentile 0.0–0.2), so ROAR reports that the removed features genuinely matter to the model. That is true. But isopropanol and methanol are not expected products of human metabolism at these levels; isopropanol is used for skin preparation, and both can enter samples during collection and handling [26], [27]. Dimethyl sulfone is less clear: it is a regular low-micromolar plasma metabolite that derives from dietary sources, intestinal bacterial metabolism and endogenous methanethiol metabolism [22], all of which could differ between a preoperative and a postoperative state. We therefore call the three compounds putatively exogenous. Their identity as handling artefacts is not established from the deposited data, and the argument below does not depend on it.

TABLE III

Case 1 replication on MTBLS881 ($n = 47$ per site). PR-AUC columns are means of per-fold average precision; Gap = model − learn-nothing scorer on held-out predictions pooled within each repeat, so it is not the difference of the two PR-AUC columns; interval from a subject bootstrap. K = number of the 11 markers required to define a responder.

Site	Outcome	Prev.	PR-AUC	Scorer PR-AUC	Gap [95% CI] (pooled)	p_{perm} (refit)	Lift	Pair-null lift med. (p95)
LA	Raw change (K=10)	0.49	0.632	0.695	−0.037 [−0.231, +0.118]	0.007	1.29	1.49 (2.09)
LA	Per-marker resid. (K=6)	0.55	0.678	0.542	+0.164 [−0.004, +0.309]	0.003	1.23	1.05 (1.25)
LA	Joint resid. (K=6)	0.43	0.619	0.545	+0.071 [−0.077, +0.190]	0.103	1.45	1.06 (1.27)
PA	Raw change (K=10)	0.64	0.757	0.723	+0.013 [−0.111, +0.154]	0.007	1.19	1.40 (1.89)
PA	Per-marker resid. (K=6)	0.51	0.631	0.546	+0.078 [−0.086, +0.222]	0.096	1.24	1.05 (1.27)
PA	Joint resid. (K=6)	0.53	0.598	0.629	−0.034 [−0.169, +0.092]	0.977	1.12	1.06 (1.23)

TABLE IV

The 21 MTBLS242 metabolites. Group: the two trios compared in Fig. 2. Direction: expected change in obesity used as a monotonic constraint (+ higher, − lower, blank = unconstrained). Step: position in the recursive ROAR removal order; blank = not removed by $k = 11$. Dimethyl sulfone, isopropanol and valine are the first three removed, so the red curve at $k = 3$ is not the same set as either trio.

Metabolite	Group	Dir.	Step
Dimethyl sulfone	Contaminant		1
Isopropanol	Contaminant		2
Methanol	Contaminant		5
L-Valine	Biology	+	3
L-Tyrosine	Biology	+	6
Lipoproteins	Biology	+	8
Glycine		−	4
D-Phenylalanine		+	7
Pyruvic acid			9
L-allo-Isoleucine		+	10
L-Leucine		+	11
L-Alanine		+	
L-Lactic acid		+	
L-Glutamine		−	
Histidine		−	
3-Hydroxybutyrate			
Acetate			
Acetoacetate			
Citrate			
Creatinine			
Hypoxanthine			

TABLE V

Spiked artefact: median rank among 22 features (1 = most important). Removal rank orders features by the PR-AUC drop when each is removed alone and the model retrained.

d	SHAP rank	Removal rank
0.75	10	14.5
1.25	3	3
1.75	1	1

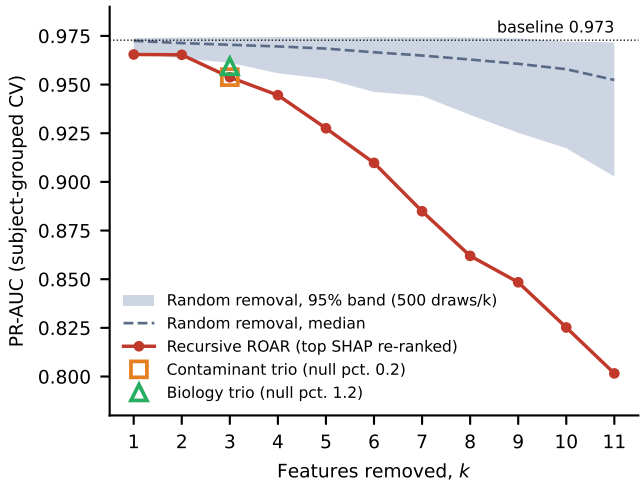


Fig. 2. Recursive ROAR on MTBLS242 against 500-draw random-removal bands. The informed curve lies below the band from $k = 3$ onward whether the removed features are putatively exogenous or endogenous.

C. Exposing control: artefact vs. biology

At $k = 3$, removing the putatively exogenous trio (dimethyl sulfone, isopropanol, methanol) gives PR-AUC 0.954 (null percentile 0.2); removing three endogenous markers (lipoproteins, L-valine, L-tyrosine) gives 0.959 (percentile 1.2). The 0.006 gap is far smaller than the fold standard deviation of about 0.03. ROAR treats the two groups alike.

A spiked positive control makes the point without relying on the identity of any real compound. We add a synthetic feature $d\,y + \mathcal{N}(0, 1)$, which depends only on the label and carries no biology, and measure both its SHAP rank and the drop in PR-AUC when it alone is removed and the model retrained (10 draws per d, Table V). At $d = 1.75$, an effect size similar to dimethyl sulfone's, the spike is ranked first by both, and removing it costs more than removing any real metabolite (0.0127 vs. 0.0073 for dimethyl sulfone). Removal-based importance ranks a feature by how much non-redundant signal it carries, whatever its origin; a low ROAR curve cannot by itself support a biological reading of the top-ranked features.

D. Caveats

This is one cohort, and its label is confounded with surgery (Case 4). Random draws overlap and share folds, so percentiles below the band mean "outside the random range," not calibrated p-values. Attribution used

all 177 samples, which makes informed removal slightly optimistic. The multi-seed ranking places L-valine above lipoproteins, the reverse of the original ranking; we kept the original endogenous trio for comparability. Both trios were chosen after the original analysis, not in advance.

VI. Case 3: Collection Protocol Equals Label

A. Setup

The Case 3 model is our original monotonic XGBoost on 132 urine spectra, evaluated by CV grouped by subject. With seed 42 it reproduces our reported PR-AUC of 0.976; over 20 CV seeds the mean is 0.911 (range 0.845–0.975), so the headline was a favourable split.

B. Why the standard check passes

Subject-grouped CV prevents the same person from appearing in training and test folds. It cannot detect a confound shared by every member of a class.

C. Exposing control: metadata and protocol audit

We cross-tabulated the label against all 64 columns of the deposited sample and assay sheets. Of these, 56 are constant (instrument, probe, pulse sequence, temperature) and six are per-sample identifiers or file paths; apart from the label, only sex varies. There is no date, run, rack or acquisition-order column. The acquisition parameters in the 132 nmrML headers are likewise constant except for per-sample calibrations (pulse width, phasing, referencing), none of which defines sessions that span both classes; pulse width alone separates the classes with ROC-AUC 0.77, which may reflect urine composition or acquisition session and cannot be resolved.

The deposited protocol, however, describes two different studies. Controls were 12 volunteers aged 18–55 with BMI 19–30, each sampled at seven fixed time points; patients were aged 30–65 with BMI 25–40 and sampled one to three times. Every control sample name begins with ADG19007u and every diabetic sample name with ADG10003u (Table VI), consistent with separate collection protocols. Age, BMI, sampling schedule, and any difference in collection, storage or diet protocol are therefore aliased with the label, and none is recorded per sample. No within-protocol comparison is possible because each protocol contains one class. The sample prefix itself is only a marker of this design (a logistic model given the prefix scores PR-AUC 1.000 under the same CV, as any perfect label proxy would); the problem is the design it reveals.

Sex is imbalanced (controls 66.7% male vs. cases 45.8% of samples) but not confounded. Subject identity is not deposited: the control numbering yields exactly 12 volunteers, matching the protocol, while the 48 diabetic samples come from 28–30 patients whose sample-to-patient membership cannot be recovered. A numbering heuristic used in our original analysis merged at least three pairs of distinct patients. Merging errs toward conservative CV, and ungrouped CV (0.969) differs little, so grouping does not drive the result; the design does, and it cannot be removed by any analysis of these data.

TABLE VI
MTBLS1 samples by label, sample-name prefix and sex. The prefix determines the label exactly; sex does not.

Label	Prefix	F	M	Samples	Subjects
Control	ADG19007u	28	56	84	12
T2DM	ADG10003u	26	22	48	28–30
Total		54	78	132	40–42

VII. Case 4: The Label Is Not What It Is Called

Our "obese vs. non-obese" task on MTBLS242 is in fact "preoperative vs. twelve months after surgery": 71 of 106 patients appear in both classes, and they supply 80% of the 177 samples. Over 10 repeats of 5-fold CV, patient-grouped CV gives ROC-AUC 0.960 and naive CV 0.959 (PR-AUC 0.973 and 0.972); the spread between repeats is about 0.006–0.009. The two agree, so the issue is not leakage but label meaning. (A single-seed 5-fold ROC-AUC of 0.973 appears in our earlier poster; it does not reproduce with the constrained model used here, which gives 0.969 under that protocol.) The sample sheet records no body-mass index, so whether any twelve-month patient is in fact non-obese cannot be checked; a year after bariatric surgery many patients need not be.

A classifier trained on this label learns everything that differs between a patient before surgery and the same patient a year later: weight loss, but also post-surgical diet, supplements and medication, fasting state, and the year-long gap in sample storage. Its score is not evidence that the serum profile identifies obesity in a general population, where lean and obese people differ in age, sex and lifestyle rather than in surgical history. The correct description is "pre- vs. post-bariatric-surgery classification," and the check that exposes the mismatch is not statistical: write down, in one sentence, what physically distinguishes the two classes.

VIII. Simulation Study

A. Design

Each subject i has a persistent level μ_{ij} for $p = 11$ outcome markers, with a shared factor (inter-marker correlation $\rho = 0.3$), and ten auxiliary features $z_i \sim \mathcal{N}(0, I)$. Observed visits are

$$x_{ij}^{(0)} = \sqrt{r}\,\mu_{ij} + \sqrt{1-r}\,\varepsilon_{ij}^{(0)}, \tag{1}$$

$$x_{ij}^{(12)} = \sqrt{r}\,(\mu_{ij} + \gamma\, g_i\, s_j) + \sqrt{1-r}\,\varepsilon_{ij}^{(12)}, \tag{2}$$

where r is the reliability (share of variance that persists between visits), $s_j \in \{\pm 1\}$ the expected direction of marker j, and $g_i = w^\top z_i / \|w\|$ a forecastable response propensity. With $\gamma = 0$ no individual response exists, so any association between baseline and outcome is regression

to the mean. The outcome is the MTBLS242-style rule "≥6 of 11 markers change in the expected direction." The model is ridge regression ($\lambda = 10$) on the 21 standardised baseline features, scored out-of-fold in 5-fold CV by average precision.

We vary $n \in \{50, 100, 200, 500\}$, $r \in \{0.25, 0.5, 0.75\}$ and $\gamma \in \{0, 0.25, 0.5\}$, with 1,000 datasets per cell. Setting $r = 0.25$ reproduces the median baseline–change correlation of -0.62 observed in MTBLS242. Each dataset is tested four ways at $\alpha = 0.05$, each with 200 null replicates and the model refitted on every replicate:

1) Label permutation on the raw-change outcome.
2) Pairing-destroyed null on lift (AP over prevalence).
3) Per-marker residualisation: within each training fold, regress change$_j$ on baseline$_j$; label from residuals; permute.
4) Joint residualisation: as (3), but regress change$_j$ on all 11 baselines.

A permutation test that holds out-of-fold scores fixed and permutes only the labels rejected 9.7% of pure-noise datasets in our sanity check; refitting on each permutation [13] gave 5.3%, so all tests refit.

B. Results

Table VII and Fig. 3 give the rejection rates.

Label permutation is wrong, and more so with more data. With no individual response, it rejected in 26% of datasets at $n = 50$ and 100% at $n = 500$ when $r = 0.25$; even at $r = 0.75$ it reached 90% at $n = 500$. More subjects estimate the regression-to-the-mean artefact more precisely; they do not remove it.

Per-marker residualisation is not enough. Residualising each marker's change on its own baseline left false-positive rates of 8–29%, again rising with n. When markers are correlated, residual change in marker j remains correlated with the baselines of the other markers, which the model also sees.

Joint residualisation is close to nominal but slightly liberal. Regressing each change on all baselines kept the rate between 5.3% and 8.1%, without a trend in n; in most cells the Wilson interval lies just above 5%, so a small excess remains. Its power when a real response exists is close to that of per-marker residualisation (e.g. 97.5% vs. 97.0% at $\gamma = 0.25$, $r = 0.25$, $n = 500$), but it is low at small n and low reliability (19% at $n = 50$ in the same setting).

The pairing-destroyed null is valid but conservative. Its false-positive rate never exceeded 4.3%, but at the MTBLS242-like reliability $r = 0.25$ and $\gamma = 0.25$ it had no power at any n (4–5%): the null's median lift matched or exceeded the model's even though a real response existed (e.g. 1.46 vs. 1.45 at $n = 500$). A model that falls below this null has therefore not been shown to lack a real response; the null detects manufactured outcomes and should not be the only test of a real one.

TABLE VII
False-positive rate (%) with no individual response ($\gamma = 0$), $\alpha = 0.05$, 1,000 datasets per row. Wilson 95% half-width ≤3.1 points.

r	n	Label perm.	Pair-null	Per-marker	Joint
0.25	50	26.1	4.3	8.0	7.4
0.25	100	52.0	2.1	9.3	6.2
0.25	200	92.6	0.7	12.0	5.9
0.25	500	100.0	0.3	18.9	6.3
0.50	50	16.3	2.2	10.9	8.1
0.50	100	35.9	0.6	13.1	6.9
0.50	200	74.4	0.2	16.3	6.8
0.50	500	99.6	0.0	28.8	6.7
0.75	50	10.6	0.9	9.6	6.4
0.75	100	18.8	0.2	13.3	6.7
0.75	200	40.4	0.0	16.2	6.5
0.75	500	90.3	0.0	27.0	5.3

C. Implication for Case 1

The per-marker correction used in our original analysis is liberal, so Section IV repeats the corrected analysis with joint residualisation. The conclusion is unchanged: the model does no better than an unfitted scorer, and the subject-bootstrap interval spans zero. Because a liberal test and a near-nominal test both found nothing, the null result is not an artefact of the per-marker correction. It remains absence of evidence at $n = 71$, not evidence that preoperative profiles cannot forecast response: the simulation shows that even joint residualisation has little power at this sample size and reliability.

IX. Discussion

A. A checklist for small-n metabolomics ML

1) If the outcome is computed from the features, do not rely on a label permutation. Residualise change on all baselines inside each training fold, and report a pairing-destroyed null as a diagnostic.
2) Refit the model inside every permutation replicate, and check that the observed statistic does not sit below the whole null, which signals a mis-specified null rather than strong evidence.
3) Include a learn-nothing baseline and report the gap with an interval that resamples subjects.
4) Audit features for exogenous compounds; report results with and without them.
5) Cross-tabulate the label against every metadata column (batch, run order, sample-name prefix, sex) and read the collection protocol for each class.
6) State in one sentence what the label physically encodes.

B. Limitations

The empirical cases use three small public datasets, chosen because their problems are verifiable from the published files, and they audit our own earlier analyses rather than a sample of the literature; we make no claim about how common these failures are. The findings

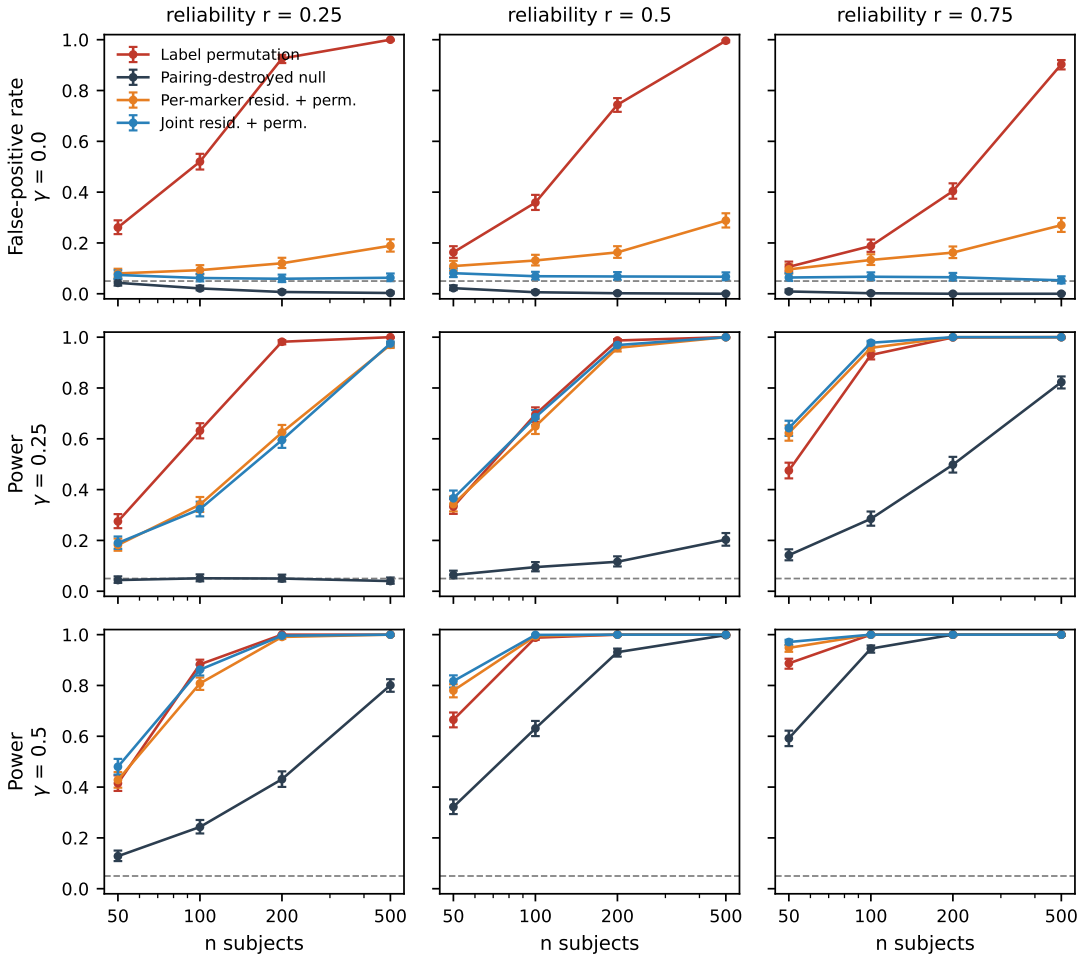


Fig. 3. Rejection rate vs. number of subjects. Top row: false-positive rate with no individual response ($\gamma = 0$). Lower rows: power with a real, forecastable response. Columns: reliability r. Dashed line: $\alpha = 0.05$. Error bars: Wilson 95% intervals over 1,000 datasets.

concern validation practice; they say nothing for or against the biology of obesity or diabetes. Regression to the mean is classical statistics, and our contribution is limited to showing how ML practice fails to detect it. The exogenous origin of the Case 2 compounds is plausible but not established from the deposited data. The simulation uses Gaussian data, a linear ridge model and one outcome rule; non-linear models and other rules may differ in degree. The pairing-destroyed null is conservative and can miss real effects at low reliability, so it should be a diagnostic, not the sole test. Finally, the joint residualisation removes any true biological dependence of change on baseline together with the artefact; separating the two needs repeated baseline measurements.

X. Conclusion

Cross-validation, permutation tests and attribution checks are necessary but answer narrow questions. In four of our own analyses of public data, a standard check passed while the result was wrong: an outcome built from its own features, putatively exogenous compounds ranked alongside endogenous metabolites, a collection protocol equal to the label, and a label named for something it did not measure. Each failure has a cheap, targeted control. We recommend that small-n metabolomics ML studies report these controls alongside their headline score, and we release code that runs them.

Data and Code Availability

MTBLS242, MTBLS1 and MTBLS881 are public through MetaboLights [4]; the datasets are not redistributed. The simulation, the Case 2 and Case 3 analyses, intermediate results and the pre-registration file for the MTBLS881 replication are in https://github.com/Calvinkub/validated-yet-wrong (tag paper-v1). Each experiment folder contains a RESULTS.md listing every reported number with its script, random seed and command. The simulation runs with python simulation/sim_pairing_null.py and takes about 11 minutes on 8 cores; –check runs the pure-noise sanity test. Cases 1 and

1b import the outcome, model and control code from our earlier mtbls242-forecast-honest repository, which must be obtained separately [19] (branch honest-evaluation, commit 637a58e). The ROAR study [20] (commit f220cd2) and the MTBLS1 notebook [21] (commit 0088e38) are the sources of Cases 2 and 3.

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